

- 1 (a) $a = \frac{F}{m} = \frac{1400}{(2600+2900+1300)}$ M1
 $= 0.206 \text{ or } 0.21 \text{ m s}^{-2}$ A1
- (b) According to conservation of momentum, the total momentum of the spacecraft and fuel is constant when no external forces acts on it. B1
 When exhaust gases are released in opposite direction to motion, the magnitude of the change in momentum of the exhaust gases is equal to the change in momentum of the spacecraft but in opposite direction. B1
 By Newton's Second Law, since the spacecraft experiences change in momentum, the rate of change of momentum is proportional to the forward thrust acting on the spacecraft. B1
**Accept the other versions. OR*
 As engine operate, they are burning fuel, which has both mass and velocity and therefore some momentum.
 By conservation of momentum, the total momemeturum of system comprising of spacecraft and exhaust engine remains constant provided no net external force acts.
 As the total mass of the system decreases due to burned fuel, the velocity of the system increases and causes the spacecraft to be propelled forward.

All underline phrases are necessary for any mark to be awarded.
- (c) The two forces are not a pair of Newton's third law M1
 Reasons A1
- The forces are of different nature
 - The forces are of different magnitude
 - The forces act on one body
- (zero or 2 marks)
- (d) There are several methods to check if the engines are able to reduce the speed of the descent module to its safe value. C1
 Actual acceleration $= F/m = 670/890 = 0.753 \text{ m s}^{-2}$
 Method 1 : Check the minimum required deceleration C1
 Method 2: Check the maximum time required
 Method 3: Check landing speed A1
 Appropriate conclusion

Mtd 1: Minimum required deceleration $= \frac{\Delta v}{\Delta t} = \frac{5.5-3.0}{3.5} = 0.714 \text{ m s}^{-2}$

(Compared to 0.753 m s^{-2}), there is sufficient deceleration, Yes

Mtd 2: Maximum time required $= \Delta t = \frac{\Delta v}{a} = \frac{5.5-3.0}{0.753} = 3.32 \text{ s}$

(Compared to 3.5 s), there is sufficient time, Yes

Mtd 3: Using $v = u + at$

$$v = 5.5 - 0.753 \times 3.5 = 2.8$$

which is safe landing speed

or using $Ft = \Delta mv$

$$670 \times 3.5 = 890 \times \Delta v$$

$$\Delta v = 2.7$$

$$\text{Landing speed} = 5.5 - 2.7 = 2.8 \text{ m s}^{-1}$$

which is safe landing speed, Yes